

Coding & Solution Report

1. Scaling Calculations

Fitted training metrics used for feature scaling:

- Mean: [36.52, 2908.91, 28.64, 381.60, 1999.84]
- Scale: [4.33, 436.58, 21.71, 147.98, 380.13]

2. Compiled Decision Tree Rules

June-September rainfall threshold: 2425.64 mm.

- If Jun-Sep Rainfall > 2425.64 mm -> Flood Risk = 1.
- Else -> Flood Risk = 0.

3. Sigmoid Logistic Confidence Score

Difference = june_sept_rain - 2425.64.

- probability = $1.0 / (1.0 + \exp(-\text{Difference} / 100.0))$.

This mathematical formula generates a smooth confidence curve without external libraries.